



NeuroSpin



ns_PULSE_pTx library

Contributors	Aurélien Massire ¹ , Redouane Jamil ² , Franck Mauconduit ² <i>¹Siemens Healthinners, Saint-Denis, France</i> <i>²Université Paris-Saclay, CEA, CNRS, BAOBAB, NeuroSpin, Gif-sur-Yvette, France</i>
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Description	This pTx pulse class helps to include external pTx pulses into sequences for Terra and Terra.X systems.
Platform	VE12U (all subversions), XA60, XB10
Contacts	Franck Mauconduit, CEA NeuroSpin

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1 Introduction

The "libns_PULSE_pTx" library contains a pTx pulse class called "ns_PULSE_pTx" that can be used to easily insert a pTx pulse into any sequence. It uses the standard Siemens framework for pTx pulses that needs an ini file. This ini file contains the gradient and RF informations, as well as the pulse duration, the nominal flip angle and several other informations.

It was initially developped to be used in the PASTeUR package which currently contains different anatomical sequences such as 3D GRE, MPRAGE, MP2RAGE and SPACE sequences. This package is also available for VE12U systems on the Siemens C2P platform [Mauconduit F (5)].

1.1 Features of the "libns_PULSE_pTx"

This library handles different features such as:

- The gradient waveforms defined in the ini file can be played in the gradient frame of reference (XYZ frame of reference) independently of the acquisition orientation. It makes it easier to tilt the volume of interest without requiring the change the ini file. A RotateGradientsPTX() method is available for this purpose.
- The flip angle can be changed by using the setFlipAngle() method. It will rescale the pTX RF pulse according the nominal flip angle available in the ini file.
- In addition, a setFlipAngleArray() mechanism is available as standard Siemens RF pulses are providing.
- The RFInfo is correctly computed to account for changes in flip angles from the nominal flip angle.
- The pulse class can provide a pointer to the RF pulse object. It can be passed on other sequence SBBs to be handled like any other RF objects.
- There is no need to define a given PPD config file.

2 Using the library

Library binaries for VE12U: The binary files must be copied into IDEA in the following locations:

```
n4
├── x86
│   └── prod
│       └── lib
│           ├── libns_PULSE_pTxLib.lib
│           └── libns_PULSE_pTxdLib.lib
└── linux
    ├── prod
    │   └── lib
    │       └── liblibns_PULSE_pTx.a
```

Library binaries for NX (XA60 or XB10): The binary files must be copied into IDEA in the following locations:

```
NX...
├── x64_r
│   ├── lib
│   │   └── libns_PULSE_pTxLib.lib
├── x64_d
│   ├── lib
│   │   └── libns_PULSE_pTxLib.lib
└── lin_r
    ├── lib
    │   └── liblibns_PULSE_pTx.a
```

Note that the debug library file contained in the NX package should be renamed from `libns_PULSE_pTxLib_debug.lib` to `libns_PULSE_pTxLib.lib`.

Makefile: The following should be added into the sequence makefile. For linux compilation, the order in which the libraries are added is important (at least in VE12U). It should be kept as follow:

```
// Required libraries for ns|_PULSE|_pTx library
STATICLIBS (libns_PULSE_pTx)
LDLIBS     (AdjUIData)
```

```
LDLIBS      (libINIAccess)
```

Once compiled, only the custom sequence binaries must be installed on the Siemens scanner. Use a static library means that there is no need to copy the library binaries at the console.

header: A class object should be added in the class definition:

```
// Adding a class object
ns_sRF_PULSE_PTX m_sRFpTx;
```

Source file inclusions: The following set of codes can be used in order to integrate a pTx RF pulse.

```
// Adding a class contract in the sequence class declaration list
MySeq::MySeq(void)
, ...
, m_sRFpTx ("PTX_PULSE_NAME") // Here, ini filename is defined
, ... // This file must be located
// in %CustomerSeq%\RFPulses\
```

Alternatively, a subfolder can be used to change the location of the pTx RF pulse:

```
// Adding a class contract in the sequence class declaration list
MySeq::MySeq(void)
, ...
, m_sRFpTx ("MySeqFolder", "PTX_PULSE_NAME")
, ... // Here, ini filename is defined
// This file must be located
// in %CustomerSeq%\RFPulses\MySeqFolder\
```

In prepare method: Preparing the object in the prepare method of the sequence. In addition, use the duration and RFInfo informations.

```
// In sequence prepare() method, prepare the object
if ( !m_sRFpTx.prep(rMrProt, rSeqExpo) )
{return SEQU_ERROR;}

// Get the pulse duration for timing purpose
lPulseDuration = m_sRFpTx.getDuration();
```

```
// Get the pulse RFInfo for SAR management
m_RFInfo += m_sRFpTx.getRFInfo();
```

Adding the object into sequence calculatePTX() to prepare the pTx pulse into the Matlab environnement:

```
NLSStatus MySeq::calculatePTX (...)
{
    // Condition to activate pTx pulse preparation such as :
    if ( rMrProt.getTXSPEC().getaPTXRFPulse().size() == 1 )
    {
        if ( !m_sRFpTx.calculatePTX(rMrProt, rSeqLim) )
        {
            UTRACE(Error,0,"Could_not_prepare_the_pTx_pulse.");
            return SEQU_ERROR;
        }
    }
    return SEQU_NORMAL;
}
```

Adding the run of the object in the runKernel of the sequence:

```
// Apply a gradient rotation of the pTx blips to be immune
// to orientation of the acquisition
if ( rMrProt.getTXSPEC().getaPTXRFPulse().size() == 1 )
{
    if ( ! m_sRFpTx.RotateGradientsPTX(m_asSLC[lSlice]->getROT_MATRIX()) )
    {
        UTRACE(Error,0,"m_sRFpTx.RotateGradientsPTX()_failed.");
        return SEQU_ERROR;
    }
}

// Run the pulse via the object using a start time
if ( ! m_sRFpTx.run( lStartTime ) )
{
    UTRACE(Error,0,"m_sRFpTx.run(_lStartTime_)_failed.");
    return SEQU_ERROR;
}
```

3 Advanced possibilities

3.1 Flip angle

By default, the nominal flip angle is used by the pTx RF pulse. However, it can be changed by the sequence, for instance by using a protocol parameter.

```
m_sRFpTx.setFlipAngle ( rMrProt.flipAngle() );
```

3.2 SAR management and flip angle of pTx pulses

The Siemens SAR management considers that an ini pTx pulse will be used with the "NominalFlipAngle" indicated in the file and SAR online monitoring will be computed with this value. If you'd like to add a flip angle parameter in your sequence, you might encounter runtime sequence abortion due to the online monitoring. To prevent this to happen, we propose to always define a "NominalFlipAngle" with a low value like 1.0. Using the class "setFlipAngle(double)" method, the sequence can use a higher flip angle than 1.0 while preventing any online monitoring abortion during sequence runtime.

In addition, to prevent any clipping of the voltages defined in the ini file, one can add the "MaximalFlipAngle" parameter in the ini file and check that the maximum flip angle is not exceeded as follow:

```
// Prevent flip angle higher than as defined in INI file
if ( rMrProt.flipAngle() > m_sRFpTx.getMaximalFlipAngle() )
{
    if ( rSeqLim.isContextPrepForMrProtUpdate() )
    {
        // This condition helps to repair the protocol if the ini file
        // has a lower maximal flip angle than the one set in the protocol
        rMrProt.flipAngle(m_sRFpTx.getMaximalFlipAngle());
    }
    else
    {return SEQU_ERROR;}
}

m_sRFpTx.setFlipAngle ( rMrProt.flipAngle() );
```

3.3 Flip angle array

The `setFlipAngleArray` method is available in this class. In this case, `setRunIndex(int)` can be used to change array index during the `runKernel`. In addition, the `getRFInfo` method is based on the standard behavior of Siemens pulses, i.e. without argument it returns a mean of the `RFInfo` of the flipangle array. It is also possible to get a given index with `RFInfo(int)`.

```
// Using a flipangle array of size 2:
double *adFA = new double[2];
adFA[0] = 5; adFA[1] = 10;
m_sRFpTx.setFlipAngleArray(adFA, 2);

// In runKernel, use the flipangle index with:
m_sRFpTx.setRunIndex(1);
m_sRFpTx.run(lStartTime); // will use the 2nd flip angle in the array
```

3.4 Pointer to pTx pulse object

In some Siemens sequences, it is useful to give the RF object address so that the pulse is completely handled by a given SBB in the sequence. The "libns_PULSE_pTx" library was prepared to handle this behaviour. For this reason, all standard RF object features were rewritten to be available in this custom pTx object. One can retrieve the pTx pulse address and then pass it on to other SBBs that will `prepare()` and `run()` the pulse as any other.

```
// Option : the IRF_PULSE object can be obtained via:
sRF_PULSE *m_pSRF;
m_pSRF = &m_sRFpTx; // Note that m_pSRF is a standard sRF_PULSE
// while m_sRFpTx is our custom ns_sRF_PULSE_PTX
```

4 Your feedback

We hope to enhance the features of this package. Your feedback is welcome regarding the use of this library. In particular, we are interested in having feedbacks on the following topics:

- if you encounter unexpected behavior, or bugs
- if you have suggestions for improvements

- if you specific features would be desirable
- if you have any tips to share

Please contact the authors of this package.

References

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- 2 Gras, V., Mauconduit, F., Vignaud, A., Amadon, A., Le Bihan, D., Stöcker, T. and Boulant, N. (2018), ‘Design of universal parallel-transmit refocusing k_t -point pulses and application to 3D T_2 -weighted imaging at 7T: Universal Pulse Design of 3D Refocusing Pulses’, *Magnetic Resonance in Medicine* **80**(1), 53–65.
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